Week 4: MAP MAKING AND MAP ELEMENTS

Politics and spatial relations in knowledge representation

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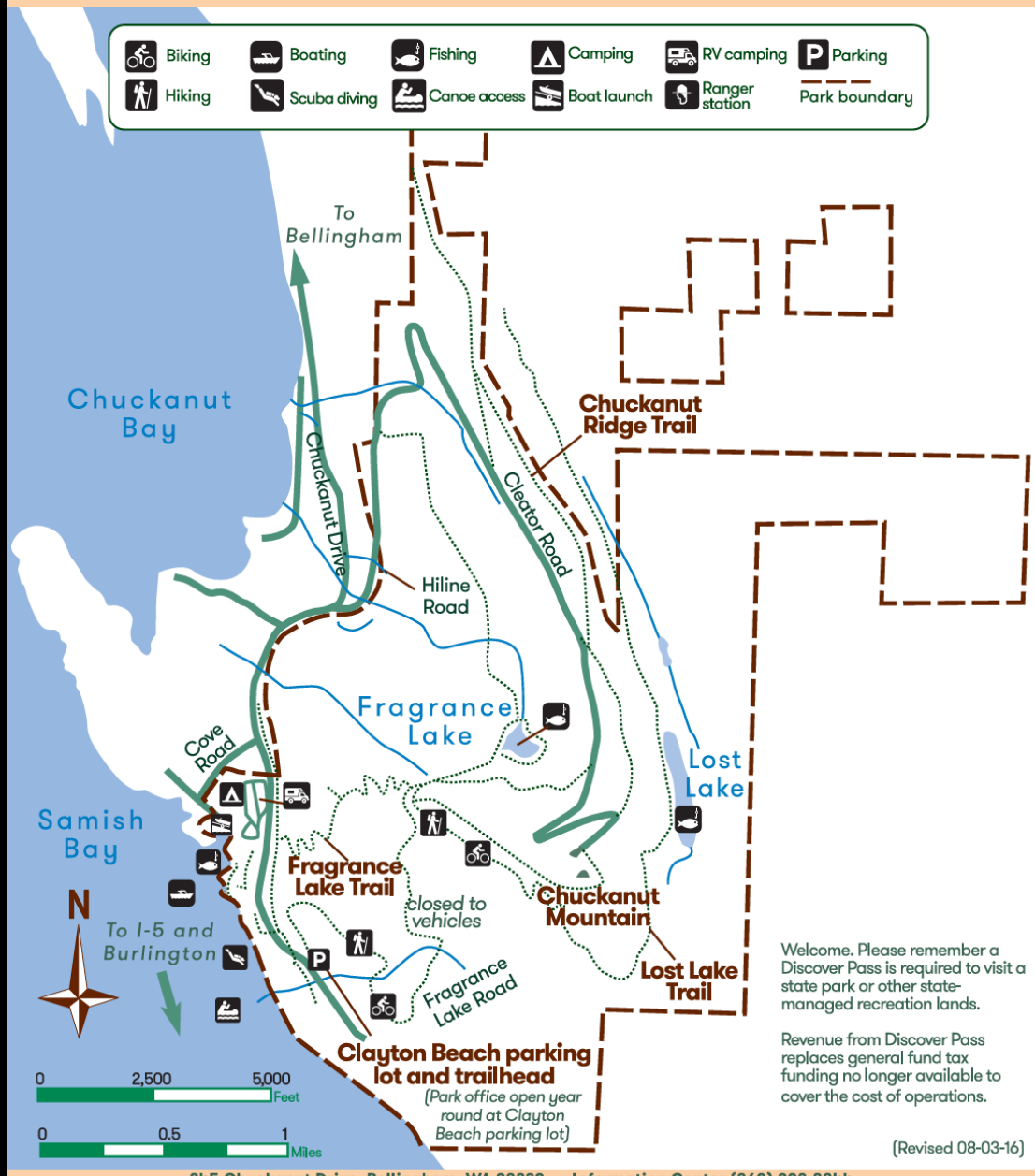
Week 4: MAP MAKING AND MAP ELEMENTS

Read:

•**KW** Chapters 4, 7, 9-12; **M** Ch2

No assignments for this week!

Larrabee State Park



Welcome. Please remember a Discover Pass is required to visit a state park or other state-managed recreation lands.

Revenue from Discover Pass replaces general fund tax funding no longer available to cover the cost of operations.

(Revised 08-03-16)

245 Chuckanut Drive, Bellingham, WA 98229 • Information Center (360) 902-8844

“Small-scale generalized maps often are authored views of a landscape or a set of spatial data. Like the author of any scholarly work or artistic creation based on reality, the conscientious map author not only examines a variety of sources but also relies on extensive experience with the information or region portrayed. Intuition and induction guide the choice of features, graphic hierarchy, and abstraction of detail. The map is as it is because the map author “knows” how it should look. This knowledge, of course, might be faulty, or the resulting graphic interpretation might differ significantly from that of another competent observer. As is often the case, both views might be valid.”

(more) map choices:

figure-ground relation
(more) symbology
map elements
…what else?

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figure-ground relation
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Figure-ground relationship

importance is highlighted

unimportant is context, background

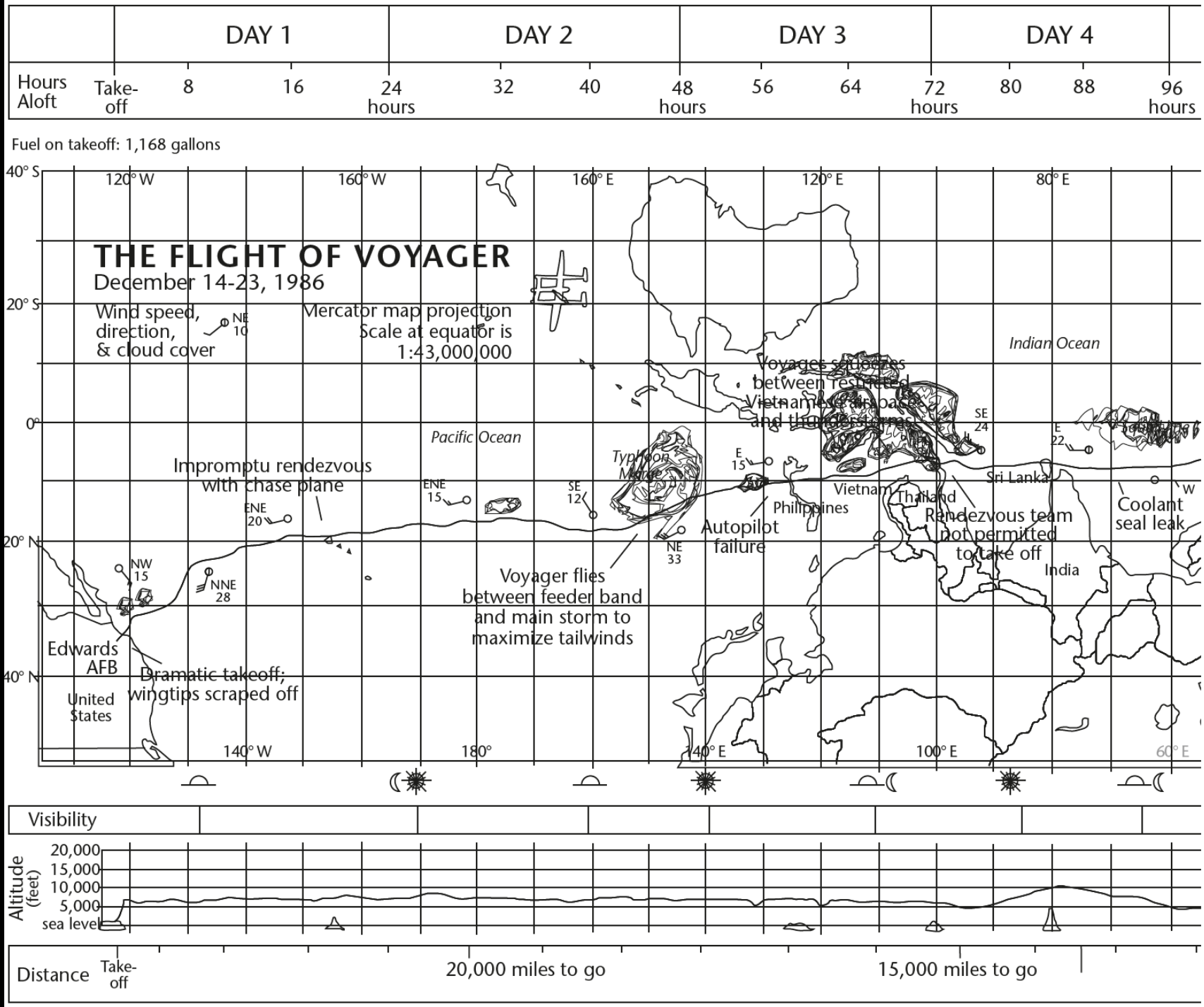

DAY 1
DAY 2
DAY 3
DAY 4
Hours Aloft
Take-off
8
16
24 hours
32
40
48 hours
56
64
72 hours
80
88
96 hours
Fuel on takeoff: 1,168 gallons
THE FLIGHT OF VOYAGER
December 14-23, 1986
Wind speed, direction, & cloud cover
NE 10
Mercator map projection
Scale at equator is 1:43,000,000
40° S
20° S
0°
20° N
40° N
120° W
160° W
160° E
120° E
80° E
140° W
180°
140° E
100° E
60° E
Pacific Ocean
Indian Ocean
Typhoon Marge
Impromptu rendezvous with chase plane
ENE 20
ENE 15
SE 12
E 15
NE 33
Autopilot failure
Voyager flies between feeder band and main storm to maximize tailwinds
Philippines
Vietnam
Thailand
Sri Lanka
India
SE 24
E 22
W
Rendezvous team not permitted to take off
Coolant seal leak
NW 15
NNE 28
Edwards AFB
Dramatic takeoff; wingtips scraped off
United States
Visibility
Altitude (feet)
20,000
15,000
10,000
5,000
sea level
Distance
Take-off
20,000 miles to go
15,000 miles to go

Figure-ground relationship

importance is highlighted

unimportant is context, background

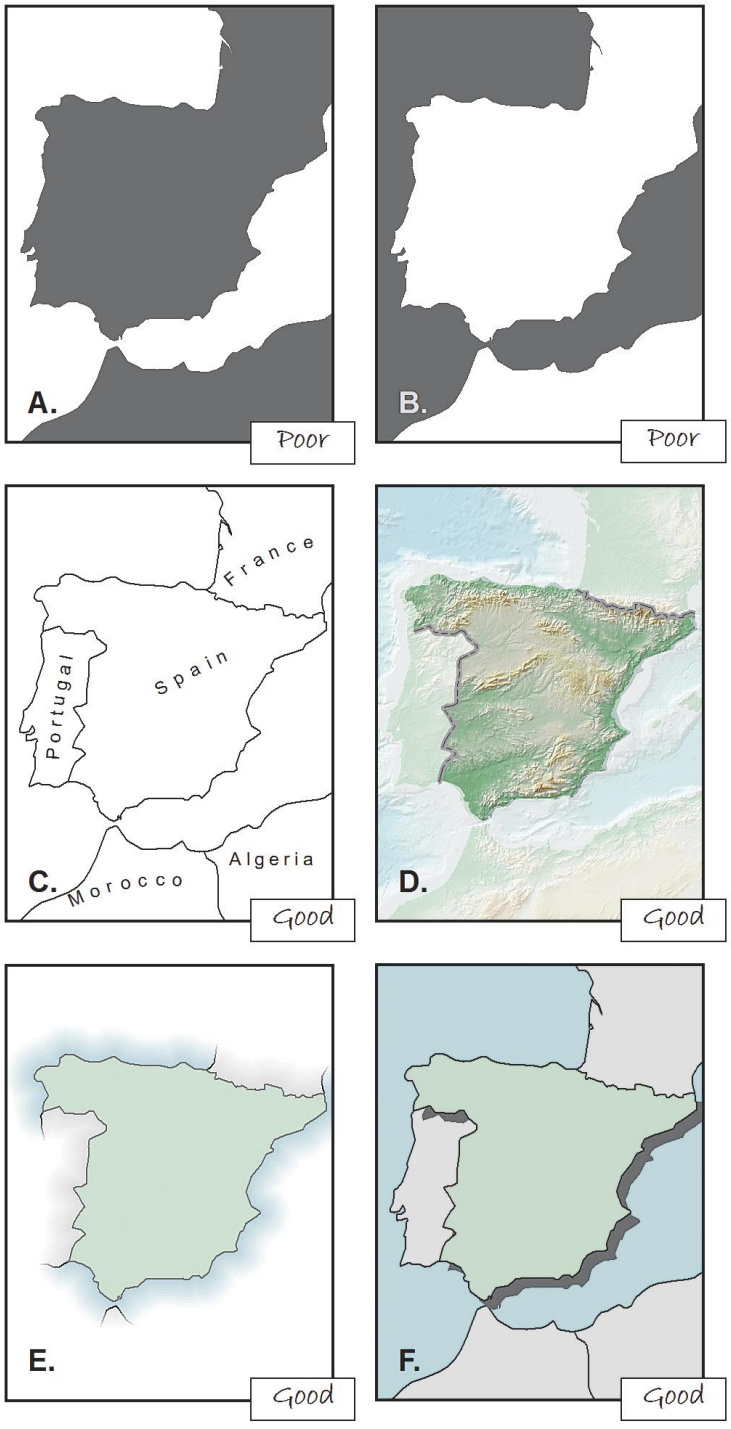
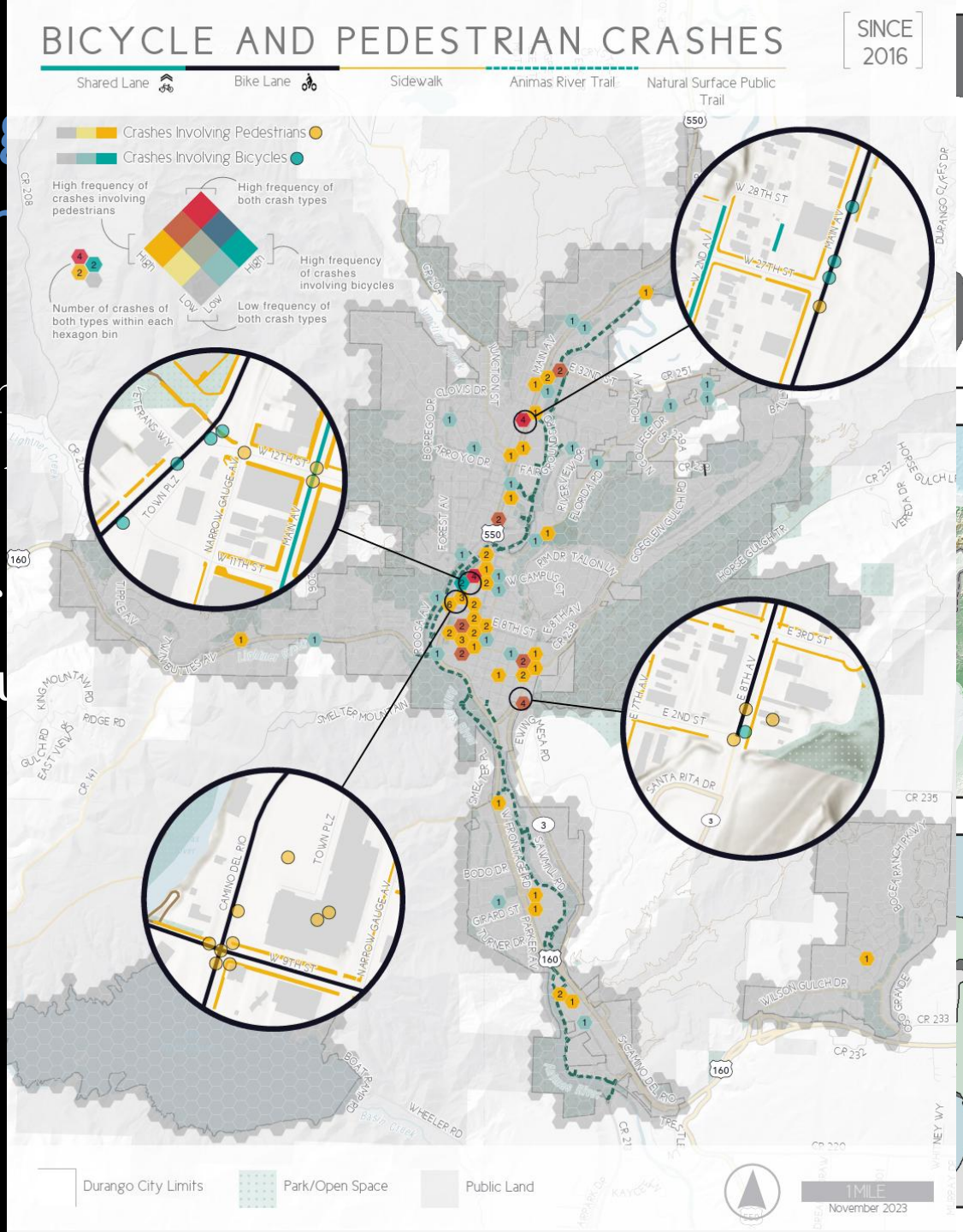
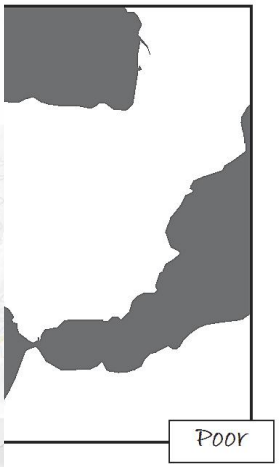


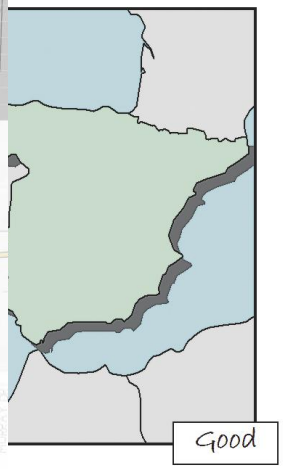
Figure-g
relation

importan
highligh

unimpor
backgrou







https://www.gisci.org/News-Old/Issues-News-Policy/gisci-celebrates-exceptional-cartographic-talent-announcing-the-2024-map-contest-winners

(more) map choices:

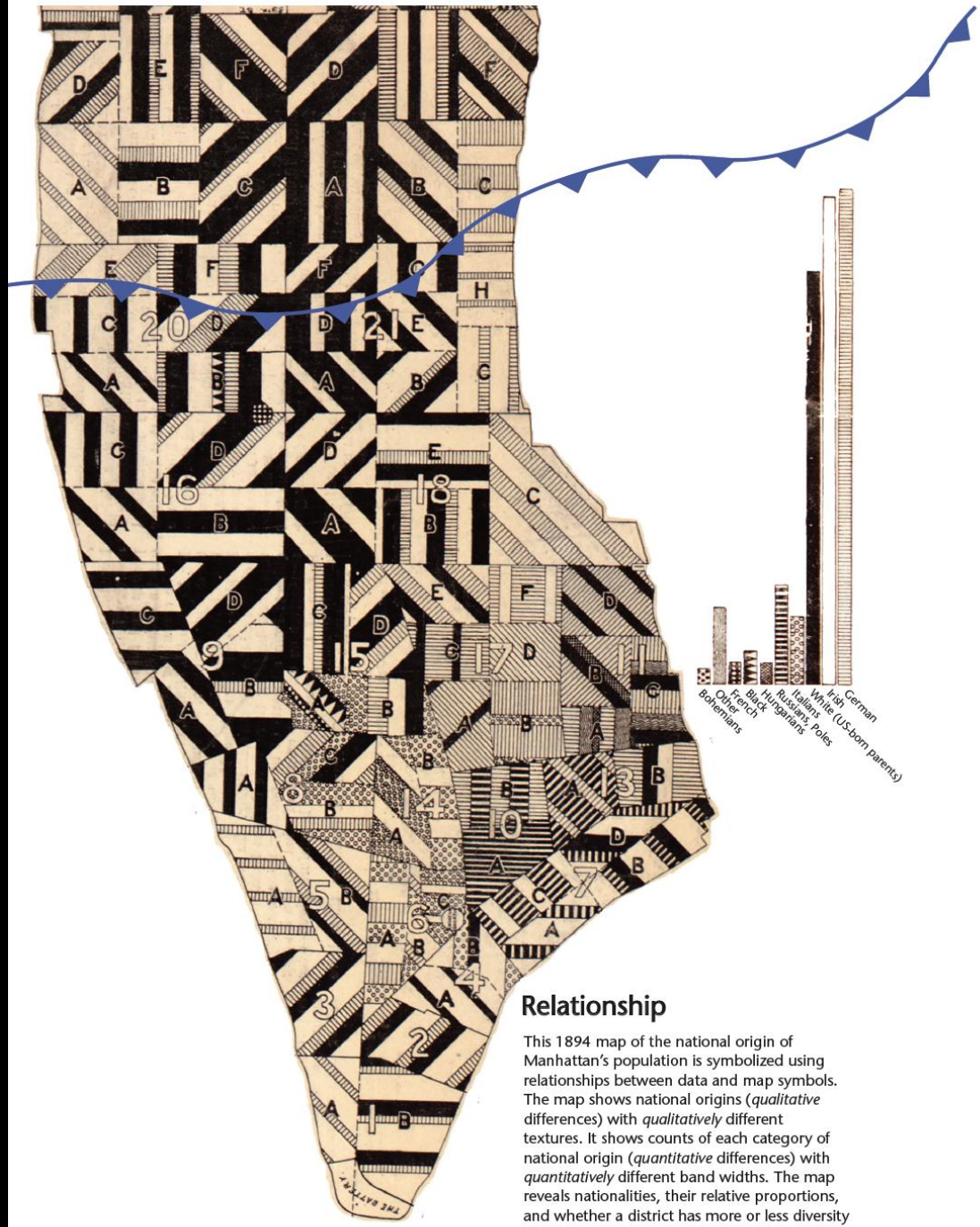
figure-ground relation

(more) symbology

map elements

…what else?

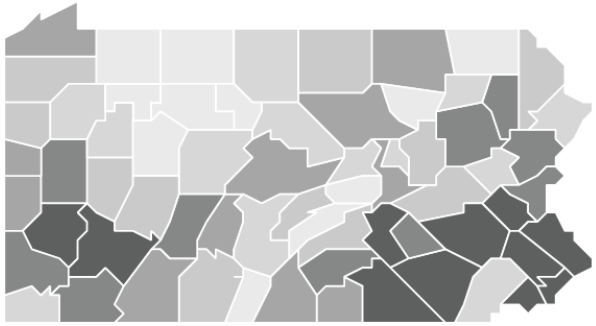
UOR TROPICAL DISEASES



Relationship

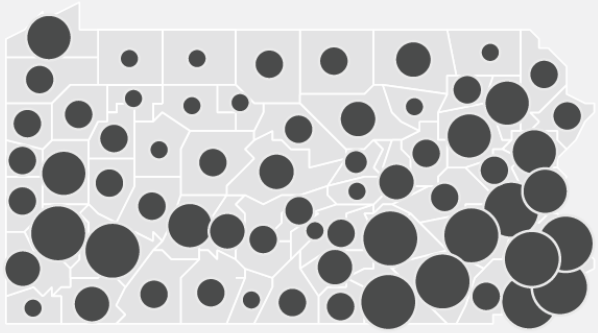
This 1894 map of the national origin of Manhattan's population is symbolized using relationships between data and map symbols. The map shows national origins (*qualitative* differences) with *qualitatively* different textures. It shows counts of each category of national origin (*quantitative* differences) with *quantitatively* different band widths. The map reveals nationalities, their relative proportions, and whether a district has more or less diversity (more or fewer bands of different textures). Symbol orientation helps distinguish areas.

Choropleth



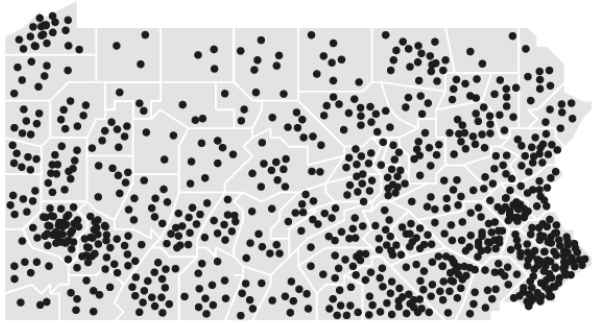
Darker means more (lighter, fewer) cases.

Graduated Symbol



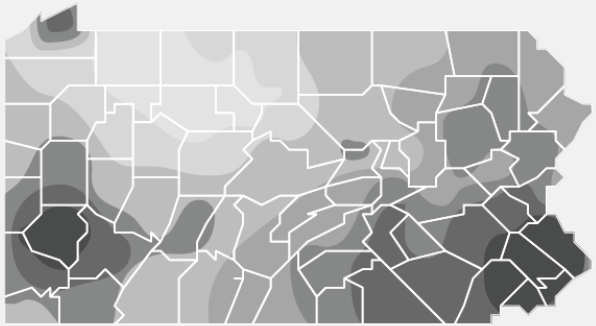
Larger circles mean more (smaller, fewer) cases.

Dot



Density of dots reveals density of cases.

Surface



Darker means more (lighter, fewer) cases.

(more) map choices:

figure-ground relation
(more) symbology
map elements
…what else?

Mali: Geography of Music & Musicians

https://www.gisci.org/News-Old/Issues-News-Policy/gisci-celebrates-exceptional-cartographic-talent-announcing-the-2024-map-contest-winners

(1) the important elements and features of a map

intellectual hierarchy

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(2) the order of their importance to the reader.

intellectual hierarchy

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(1) the important elements and features of a map

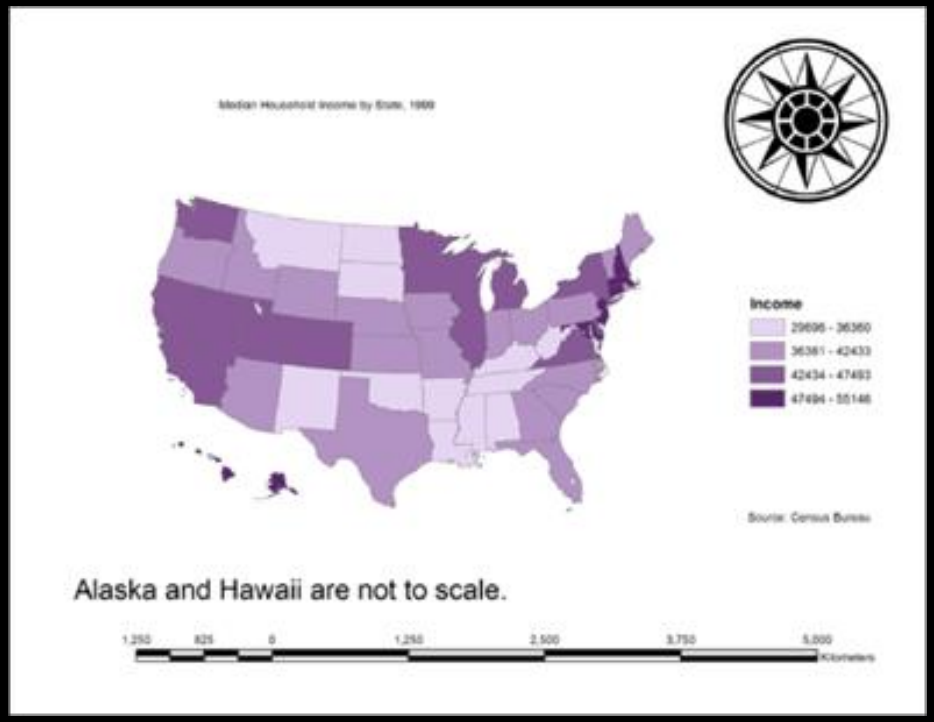
(2) the order of their importance to the reader.

i.e.:

1. population density in Seattle
2. roads
3. contextual information (the state, Van is., etc.)
4. map title
5. subtitle
6. legend
7. scale bar
8. N arrow
9. projection, data source, author, etc.

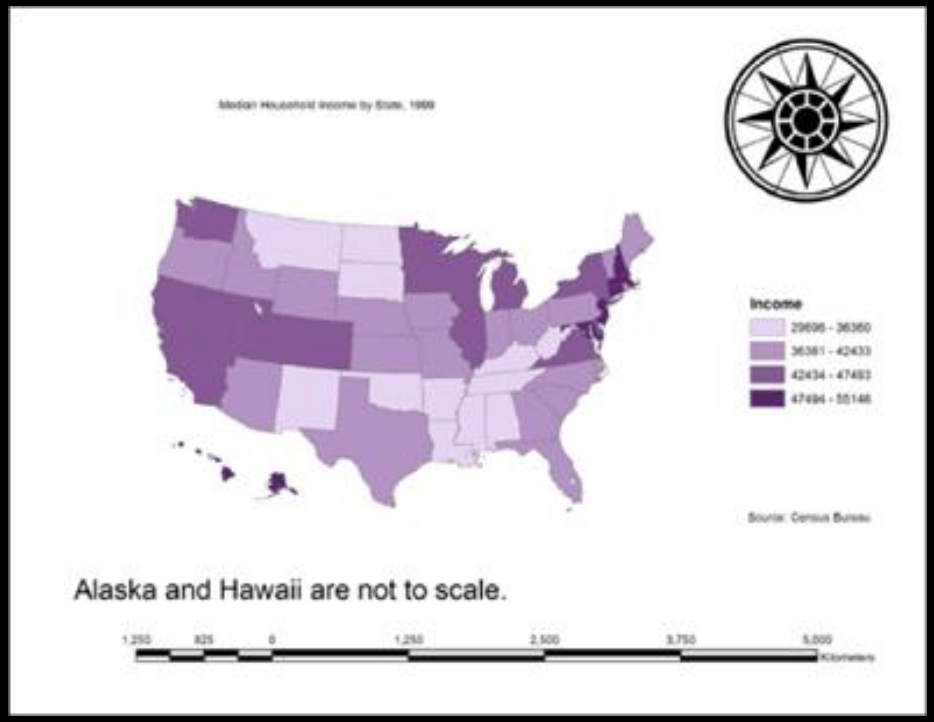
reinforces the intellectual hierarchy

visual hierarchy



intellectual hierarchy

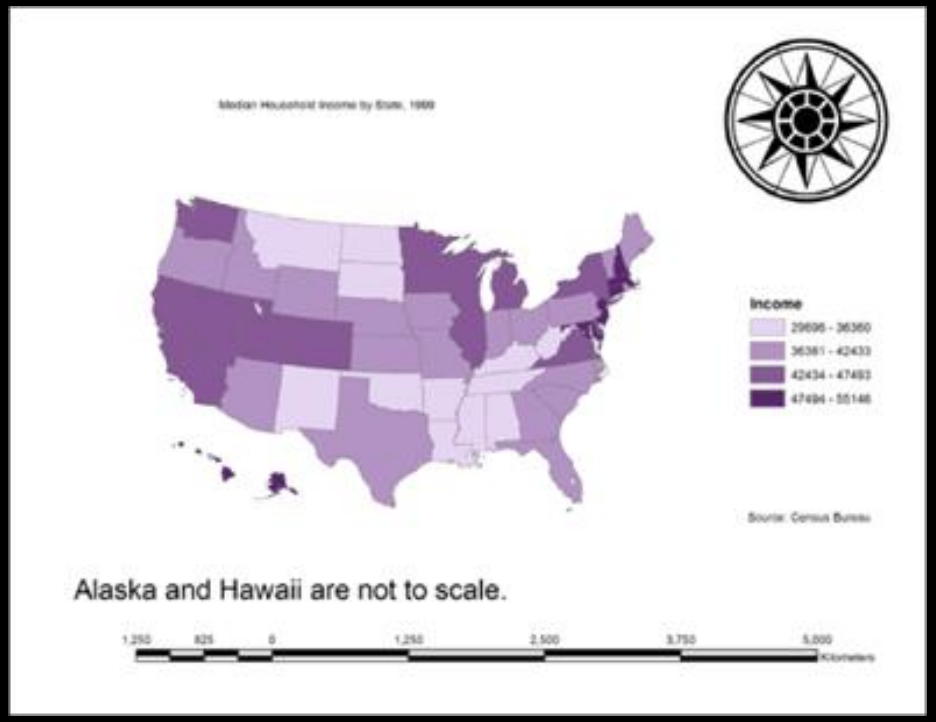
visual hierarchy

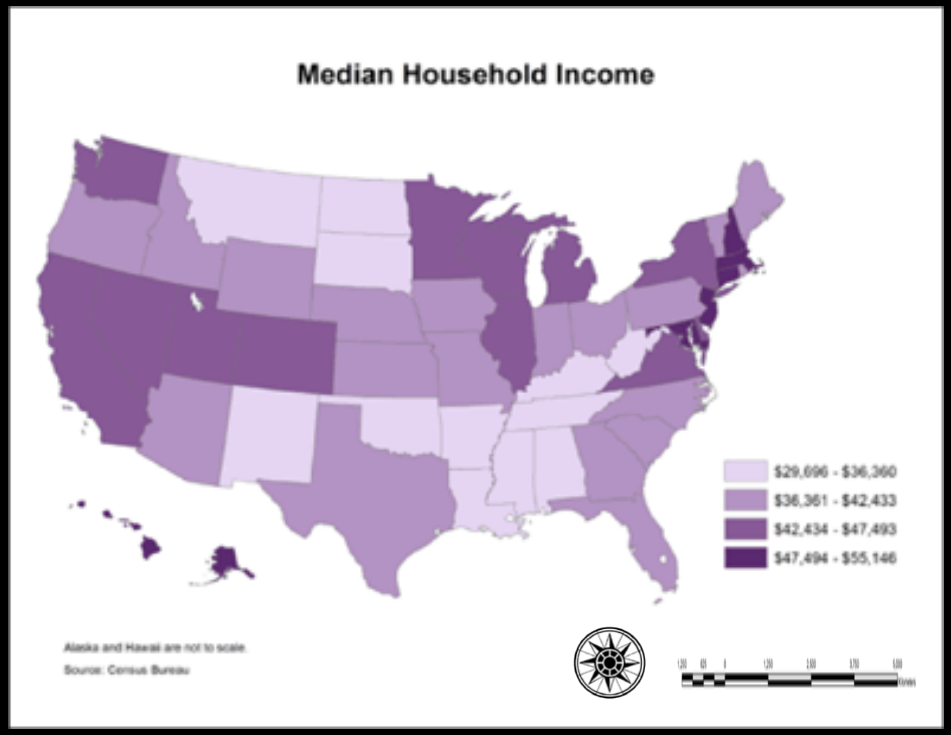


https://www.e-education.psu.edu/geog486/node/1858

intellectual hierarchy

visual hierarchy





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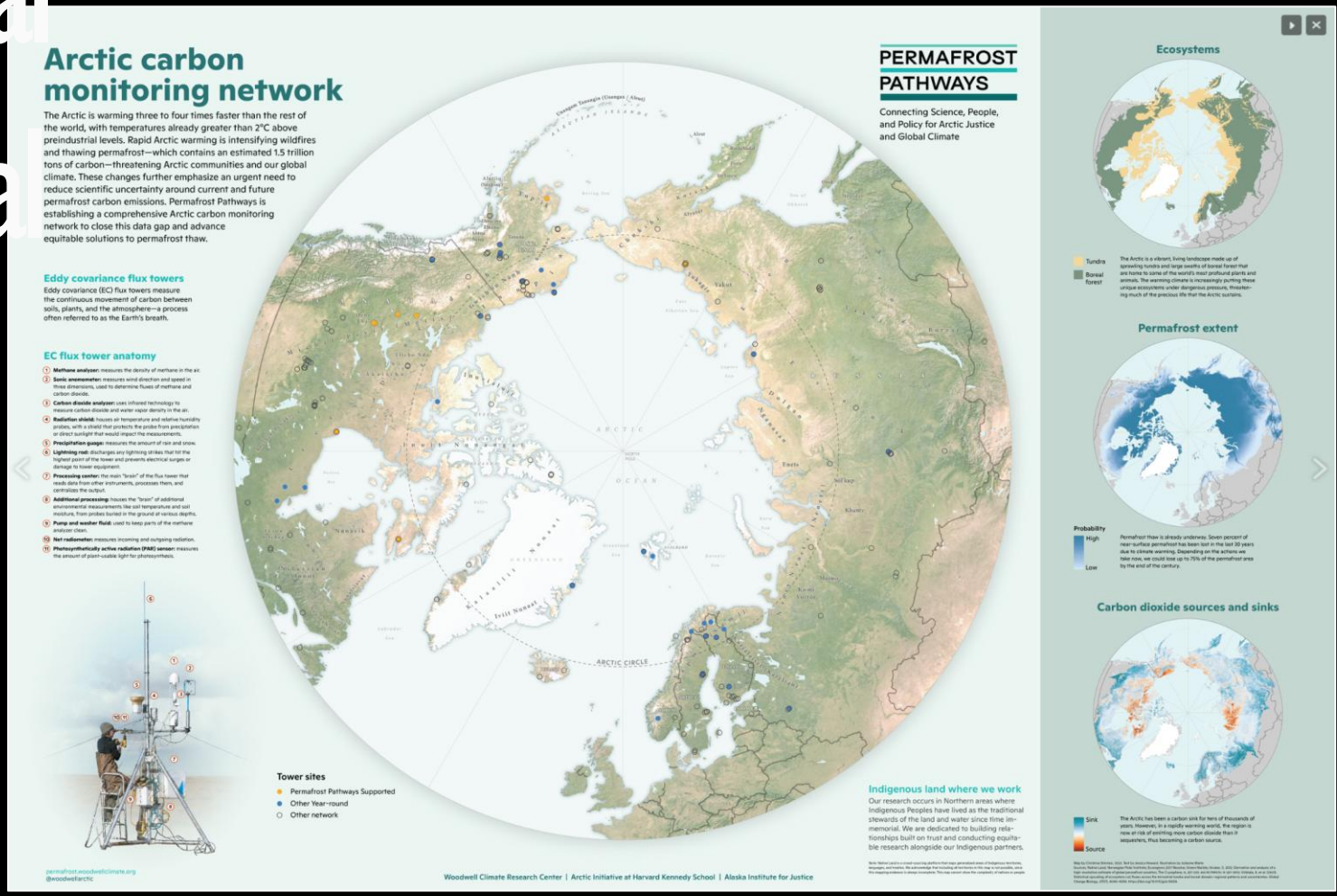
intellectual and visual hierarchy

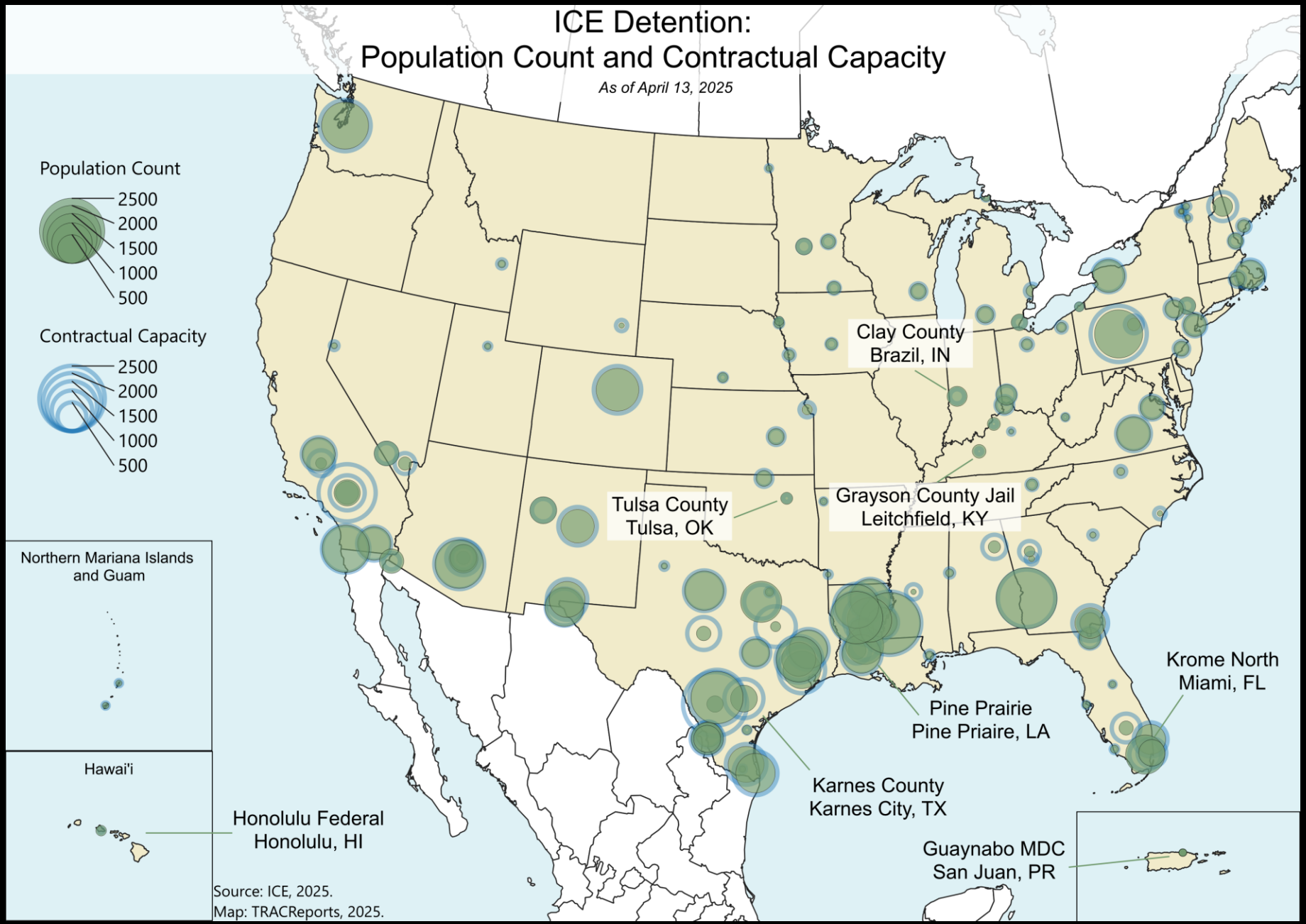


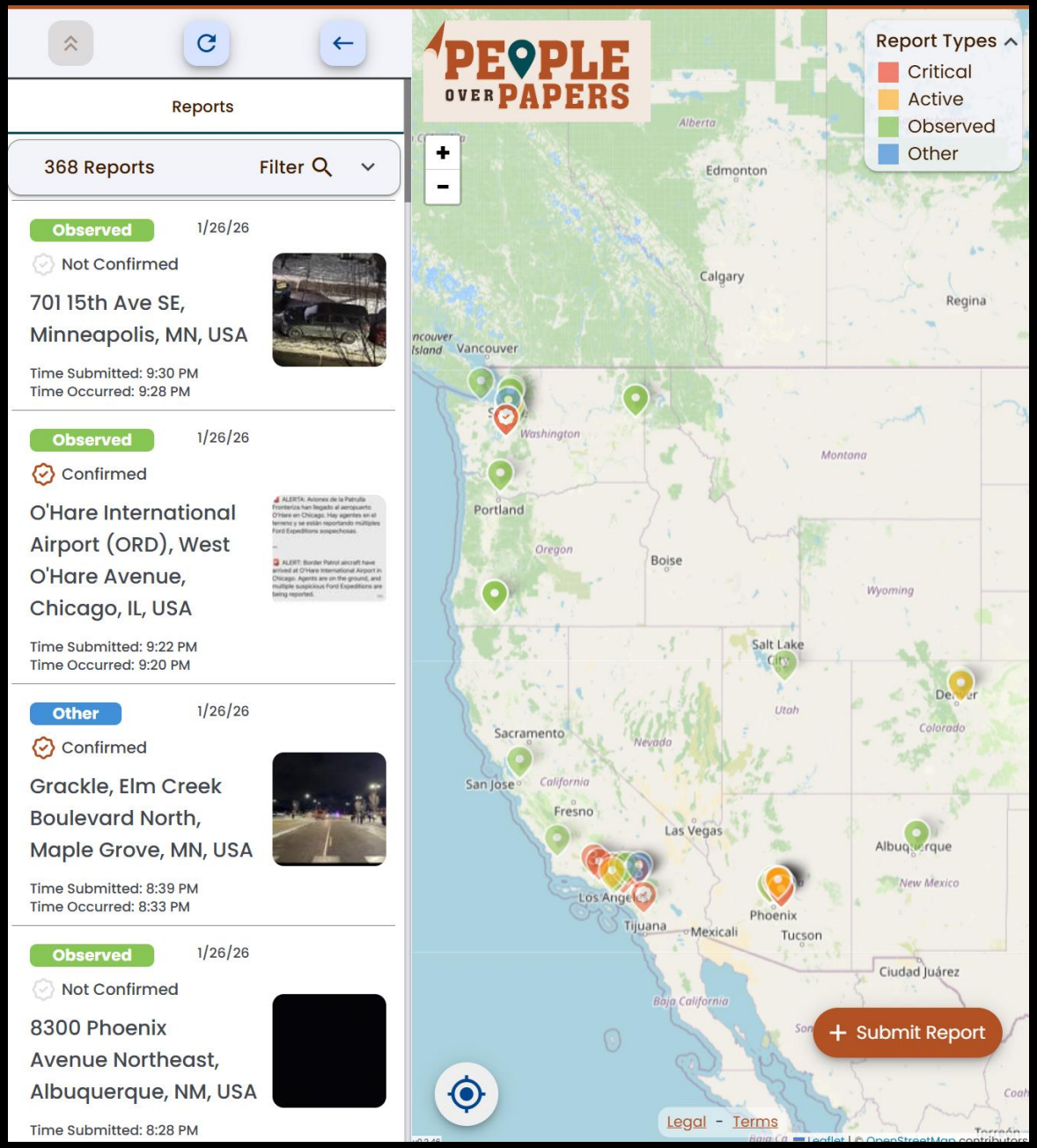
is a map necessary or useful? if you think so, what would be necessary to add here?

https://www.reddit.com/r/MapPorn/comments/1qlxxdb/8_us_states_have_laws_that_prohibit_atheists_from/

power of maps [pt 1]

https://tracreports.org/reports/762/



power of maps [pt 2]

https://iceout.org/en/